

The Emergence of Animacy Through Time in Compositional Pattern Producing Networks

Alex Alvarez

Independent Researcher, Boston, MA, USA
calvarez1@tamu.edu

Abstract

Picbreeder showed that human-guided evolution over compositional pattern producing networks (CPPNs) can discover surprisingly meaningful images in an otherwise vast design space. This paper introduces GIFBreeder, a Picbreeder-style extension in which CPPNs are given an explicit time input and therefore define short videos rather than static images. The motivating observation is qualitative but striking: when a human explores this space interactively, some evolved clips appear creature-like, as if small agents were swimming through a field. I ask whether these impressions reflect stable statistical properties of the resulting videos and whether those properties can be related to the exact programs encoded by the CPPNs. These results support a modest but important claim: human-guided search appears to favor a tractable regime in the space of analytic spacetime programs: relatively uniform canvases populated by a small number of localized, temporally coherent entities. Exact program export and node-level inspection show how such entities can be produced by compact subprograms, including time-gated Gaussian envelopes that selectively darken or recolor localized regions. GIFBreeder therefore offers a new artificial life platform for studying how human preference, generative encodings, and apparent animacy interact in open-ended exploration.

Submission type: **Full Paper**

Data/Code available at: https://github.com/calvarez0/gifbreeder_alife

Introduction

The space of all possible three-second videos is combinatorially enormous. Most artifacts in that space are visually meaningless, and a random sample almost always results in complete noise (Figure 1a). However, in some remote corner of this vast and desolate search space exists a clip from your favorite movie, video of the first human landing on the moon, and, potentially, incredible undiscovered art. In Picbreeder, users evolve images by selecting preferred outputs and allowing a branching population of compositional pattern producing networks (CPPNs) to mutate and

recombine (Secretan et al., 2011; Stanley, 2007). Similar ideas have been explored in interactive evolutionary computation more broadly (Takagi, 2001) and in earlier animation-oriented work such as Karl Sims’ evolutionary graphics systems (Sims, 1991). More recently it has been shown that time-indexed interactive evolution can support the breeding of 2D and 3D animations as well as static images (Tweraser, et al., 2018).

This paper studies what happens when a Picbreeder-like system is given an explicit time input and is used to evolve short animated GIFs. The resulting system, GIFBreeder, produces 3-second, 15 fps clips by querying a CPPN over image coordinates and time. GIFBreeder was developed as an experimental tool for evolutionary art. However, in exploring its output space, we began to observe the emergence of strikingly lifelike motifs. Crucially, these phenomena do not arise from simulated agents or physical dynamics: GIFBreeder does not implement a world model or iterative update rules. Instead, each animation is generated as a direct function over space and time. In that sense, each phenotype is better understood as a compact spacetime field rather than as a physics simulation.

The motivating empirical observation is that some clips nevertheless appear animate. Localized dark or colored forms seem to approach one another, veer, regroup, or swim through a bright field (Figure 2). These clips evoke a long line of work in vision science and cognitive science showing that apparent animacy can arise from motion alone, even in simple geometric displays (Heider and Simmel, 1944; Scholl and Tremoulet, 2000; Tremoulet and Feldman, 2000; Gao et al., 2009). They also invite comparison to the “bug detector” results of Lettvin et al. (1959), in which specific motion and contrast statistics drive strong responses in biological visual systems.

The question is not whether GIFBreeder has produced literal organisms, but whether human-guided search through CPPN spatiotemporal space can discover expressions whose statistics fall into regimes that human perceptual systems interpret as animate. If so, GIFBreeder could become an unusual artificial life platform: one in which apparent lifelike-

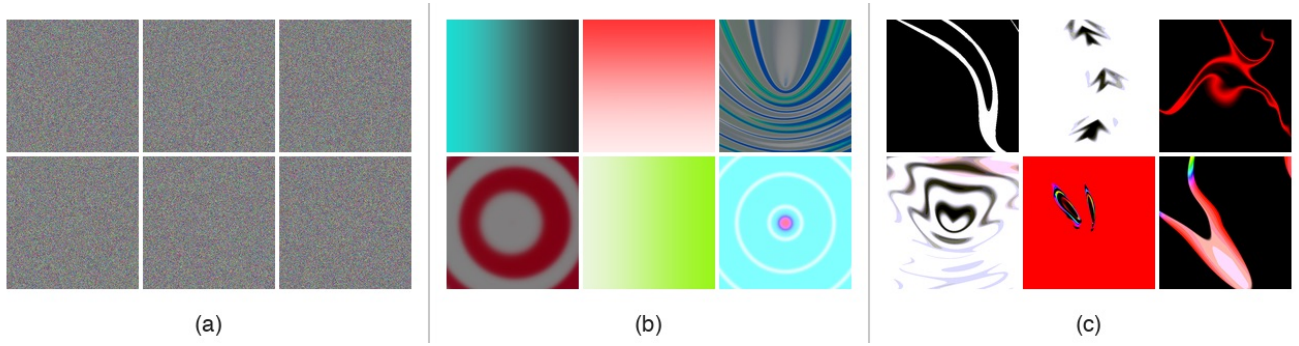


Figure 1: Middle frames of 3-second GIFs generated by three methods. (a) Pure random noise (independent per-pixel, per-frame RGB). (b) Random CPPNs (4–20 hidden nodes, random weights and activations). (c) CPPNs evolved via interactive human selection in GIFBreeder. Human aesthetic pressure rapidly produces structured, recognizable spatiotemporal patterns absent from both random baselines.

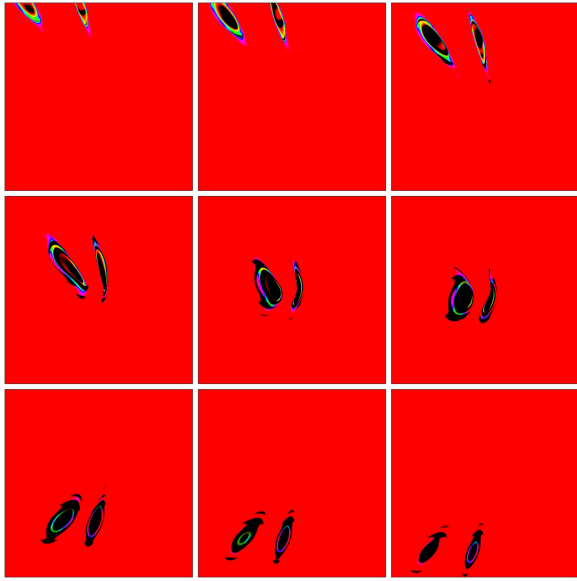


Figure 2: A GIFBreeder expression shows two limbomorphs appearing from the top left before colliding into one another, redirecting their trajectory, and continuing their journey together. This is consistent with behavior that one would infer belonged only to living beings, invoking apparent animacy (Heider and Simmel, 1944).

ness emerges not from iterated local rules, as in cellular automata or Lenia-like systems (Wolfram, 1984; Chan, 2019), but from direct parameterizations of spacetime fields.

This paper makes three contributions:

1. It introduces GIFBreeder as a Picbreeder-style system for interactively evolving animated GIFs from CPPNs with an added input: time.
2. It reports a first controlled animate-versus-inanimate analysis on a labeled corpus of saved GIFBreeder genomes.

3. It argues that the most robust statistical signature of perceived animacy in this corpus is a figure-ground organization: relatively homogeneous backgrounds with a small number of localized, temporally coherent entities.

The current evidence does not warrant saying that GIFBreeder discovers genuine agents or local reaction-diffusion-like dynamics. What it does suggest is that interactive evolution, guided by a human evaluator, can find a region of spacetime-function space that sits near the perceptual triggers for animacy.

Platform Methods

GIFBreeder is a web-based interactive evolutionary platform inspired by Picbreeder that uses NEAT-based search over CPPN representations to evolve animated GIFs through user-guided selection. The current implementation uses a population size of 20, up to 8 hidden nodes for each the initial random CPPN structure, and standard Picbreeder-style structural mutation and evolution over nodes, connections, weights, and activation functions. The chosen CPPN inputs are x , y , radial distance from center $[(d = \text{sqrt}(X^2 + y^2))]$, time t , and a bias constant b . X and Y inputs are normalized between -0.5 and 0.5 while t is normalized between 0 and 1 . The three outputs are interpreted and displayed as HSV channels but can be converted to RGB.

Formally, a genome parameterizes a CPPN

$$F_{\theta}(x, y, d, t, b) \rightarrow (h, s, v),$$

where θ denotes the network topology, activation functions, and connection weights, x and y are pixel coordinates, t is the current time step, and b is the bias node (always 1).

A phenotype is obtained by querying F_{θ} over a pixel grid and $t = 45$ timesteps, corresponding to 3 seconds at 15 frames per second. In the web-app implementation they are presented to the user as 64×64 spatial resolution renderings but in figures reported here, clips were rendered at $512 \times$

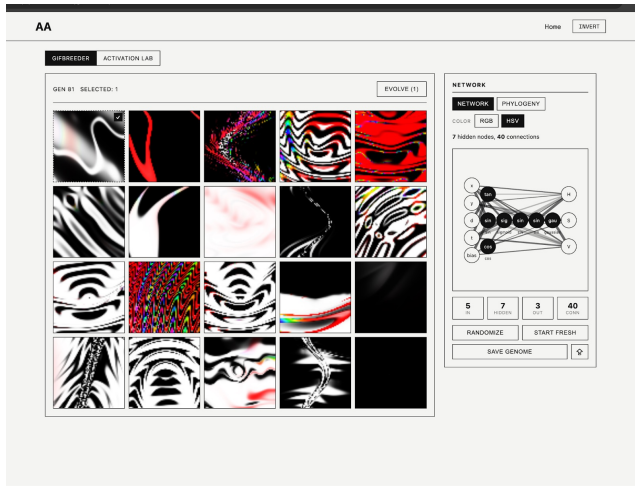


Figure 3: The GIFBreeder interface showing a population of 20 archived individuals on the left. The CPPN of the first (top left) expression is shown in the sidebar on the right.

512 spatial resolution for analysis. Because the phenotype is generated by normalized coordinate queries rather than a recurrent update rule, GIFBreeder is resolution invariant. GIFBreeder defines a spacetime field rather than a simulated world. This distinction is central to the interpretation of the results below.

The set of activation functions used by CPPNs in GIFBreeder are sigmoid, tanh, Gaussian, sine, cosine, absolute value, ReLU, and identity. Some of these were adapted from the original picbreeder and others were added for experimental purposes. The kinds of activation functions chosen can bias the output and thus the activation functions were chosen to reflect nature. Sine and cosine were chosen as oscillation functions because wave-like, repeating spatial forms are common in nature (Gray, 1946; Turing, 1952; Murray, 1981; Kondo and Miura, 2010). The Gaussian was chosen for its localized structure properties, allowing for blur, diffusion, and noise distribution (French and Brakefield, 1992; Monteiro, et al., 2001). Sigmoid and tanh were chosen for their ability to represent smooth saturation and soft interfaces (Timoney, 1993). ReLU was chosen for its thresholding capacity (Melen, et al., 2005; Skopelitis, et al., 2017). The absolute value and identity functions can be thought of as building blocks that help reproduce the symmetries and local linearities common in nature (Holló, 2015). All other functions and combinations of functions in the CPPN can also be thought of as stepping stones or building blocks for more complex composite functions and ultimately their expressed phenotypes. The NEAT algorithm is used for search, just like in the original picbreeder, because of its ability to elaborate and complexify open-endedly (Stanley and Miikkulainen, 2002; Stanley and Miikkulainen, 2004). The exact implementation and

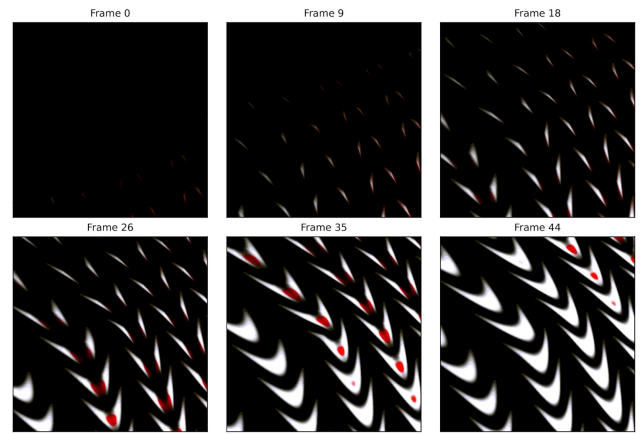


Figure 4: Six equidistant frames of a GIFBreeder expression. If the user only saw the first frame, they would miss the “butterfly formation” flowing from the bottom right.

hyperparameters including mutation rates, weight perturbation rates, and initialization methods, can be found in the source code.

Breeding

For each generation, the user sees a population of N animated GIFs. The user can select M individuals as parents for the next generation, where $1 \leq M \leq N$. After selection, M chosen parents are copied unchanged into the next population as pure elitist selection. The remaining $N-M$ slots are filled with offspring until the population contains N individuals again. For each offspring, we sample two parents independently and uniformly from the selection set with replacement (the second draw does not remove the first parent from the pool). If the same parent is sampled twice, which is always the case when only one individual is chosen to evolve, the offspring is produced as a mutated clone of the first sampled parent. Otherwise, the two parents are crossed over and the resulting child is then mutated. When more than two parents are selected, GIFBreeder does not perform a many-parent recombination over all selected individuals; instead, each child is derived from only one or two randomly sampled selected parents.

Results

Randomized pixels over 45 frames show noisy images, while randomized GIFBreeder CPPNs show more structure but nothing meaningful (Figure 1a-b). However, GIFBreeder CPPNs evolved interactively by human selection seem to quickly create recognizable spatiotemporal patterns (Figure 1c).

Talk about how giving a time variable allows a CPPN genome to come to life, and if there wasn’t time, we would simply miss out on certain expressions and art that only time could reveal (Figure 4).

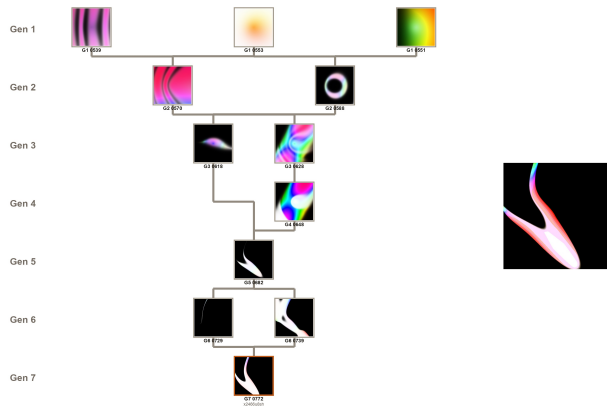


Figure 5: A GIFBreeder individual (x2466u8sh) evolved apparent animacy in only 7 generations. Phylogenetic tree on the left and frame 18 of the phenotype on the right.

Limbomorphs

Lifelike beings can sometimes emerge from within these GIFBreeder phenotypes. These phenomenological beings, which we'll call limbomorphs because they live in a liminal three-second world, can begin to emerge as primitive animacy in as little as 7 generations and possibly fewer (Figure 5).

Talk about how they can complexify over time (Figure 6) and have phylogenetic families (like figure 6) but also mention the red background black blob family (which we can see one individual in figure 2).

Conclusion

GIFBreeder is a new form of computational media, extended from Picbreeder, initially created as a tool for evolutionary art but discovered to create lifelike beings that emerge purely under human aesthetic selection. Adding a time dimension to Picbreeder creates a new search space of compact space-time fields which can bring images to life. By allowing humans to guide the evolutionary process of these expressions, they implicitly imbue the CPPNs with what they find most salient and interesting. Sometimes the result of this process is emergent lifelike beings, limbomorphs, which can tell us both about the nature of living things and about the cognitive dispositions of what humans find interesting or salient. This can be used to reverse engineer the appearance of life and also reveals something about the user's cognitive dispositions. Furthermore, GIFBreeder can be used as a new tool for evolutionary art and artificial life media.

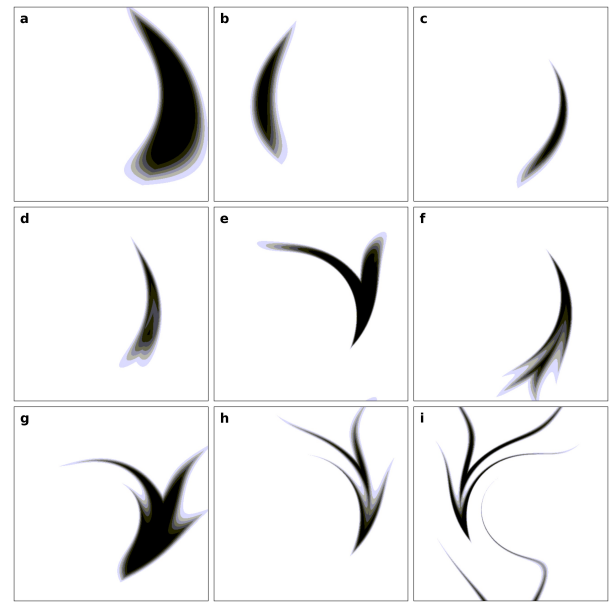


Figure 6: Nine limbomorphs of increasing morphological complexity. They begin as a large undifferentiated mass (4a-b) before becoming thinner (4c), developing a forked tail (4d-e), a secondary bifurcation (4f-g), and eventually fine, filamentous (flagella-like) appendages (4h-i).

Acknowledgements

I thank the creators of Picbreeder and related interactive evolutionary systems for establishing the conceptual and technical foundations upon which GIFBreeder stands.

References

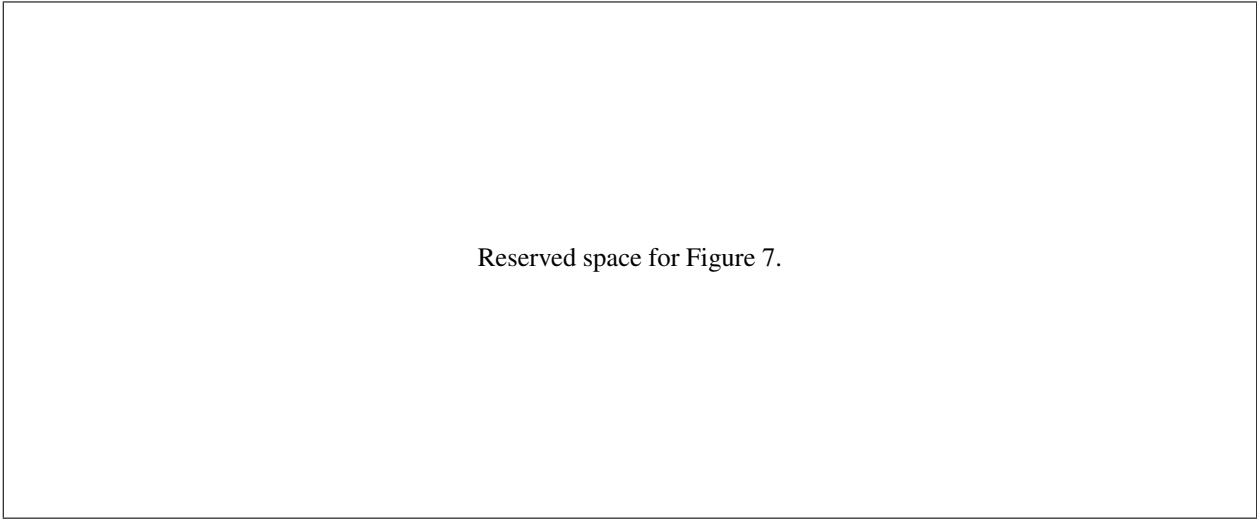


Figure 7: Reserved space for Figure 7.

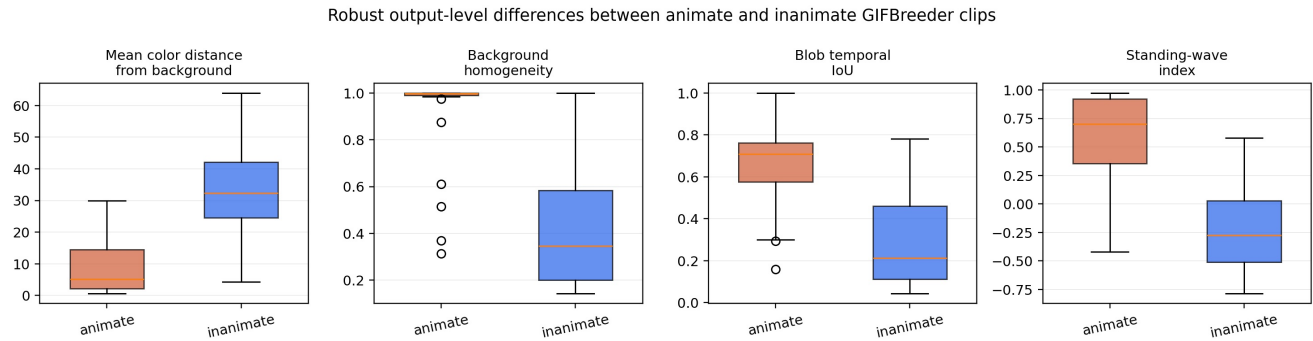


Figure 8: Four output-space measurements that most clearly distinguish the animate and inanimate corpora. These metrics are correlated, but they all tell the same story: animate clips tend to consist of localized, coherent entities on top of relatively homogeneous backgrounds.